Riteng Zhang

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Education

Boston College — *Sep 2020 – May 2024*

B.A in Computer Science
B.A in Mathematics
Minor in Philosophy

Research Interest

- Mechanistic Interpretability
- Knowledge Extraction from ML Models; AI for Science
- Neural Architecture Search/Neuro Inspired AI
- Transparent, Trustworthy, Robust AI; ML Safety

Awards & Achievements

John McCarthy, S.J., Award for Natural Science — *Boston College, May 2024*
Awarded annually to one student for the best senior thesis in Natural Science, judged by dean's reviewers and administered by the Morrissey College of Arts and Sciences Dean's Office

Neuhauser Award — *Boston College, May 2024*
Awarded annually to one student in the Computer Science department who achieves the most recognition from all CS faculty

Scholar of the College — *Boston College, May 2024*
Awarded to exceptional students who have excelled academically and completed substantial, high-quality independent work under faculty supervision during their senior year

Advanced Study Grant — *Boston College, April 2023*
Awarded for exceptional research proposals to support advanced study

Research Experience

Interpretability Formalizing and Auto-Explaining Framework — *Boston, April 2023 – May 2024*
Honors Thesis, advised by Professor Emily Prud'hommeaux

- Composed a survey of deep learning interpretability methods and taxonomies
- Built toolkit to codify deep learning elements, enabling automated interpretability method taxonomy
- Developed automated implementation pipeline for interpretability methods
- Created LLM-powered interface for adaptive interpretability method exploration given context (e.g modality/model defined)
- Awarded John McCarthy S.J Award

Analyzing Micro-Level Structural Modularity for Enhanced Interpretability — *Boston, July 2023 – Oct 2024*
Independent Study, advised by Professor Sergio Alvarez

- Conducted comparative analysis between monolithic (e.g model with varying attention head sizes, or inception-like branches) and non-monolithic architectures to investigate how structural modularity influences functional modularity and interpretability

Yuan's Lab - Self-Supervised Multimodal Representation Learning *Boston, Sep 2024 – Present*
Team Lead, in Professor Yuan Yuan's Lab

- **Leading a team of 5 researchers** and systematically designing and experimenting with variants of Concept-Encoder Diffusion models, DINO, SimCLR, and Barlow Twins for **disentangled representation learning**
- Designed an attention-based interpretability tool for cross-modal correlation retrieval in Stable Diffusion and its variants
- Test the framework on science-related datasets, such as 3D MRI and ADHD-related textual data pairs from the ABCD dataset

Language Models Evaluation for Neuroatypical Language *Boston, Jan 2023 – Jan 2024*
Research Assistant for Professor Emily Prud'hommeaux

Large ASR Model Evaluation for Under-resourced Language *Boston, Jan 2023 – Sept 2023*
Research Assistant for Professor Emily Prud'hommeaux

Evaluation of LLM Few-Shot Ability when Expecting Formatted Output *Boston, Nov 2022 – April 2023*
Independent Study, advised by Professor Emily Prud'hommeaux

CS Placement Test Automation Framework *Boston, May 2023 – Aug 2023*
Blossoms AI Research, advised by Professor Maira Samary

Work Experience

Blossoms AI *Boston, May 2023 – Present*
Co-founder and Chief AI Officer

- Blossoms AI is an education technology company with a team of 15 employees. We specialize in developing HCI/AI Apps designed to assist with personalized and customized teaching.
- **Products deployed in several colleges** in the Boston area, and in a New York high school district
- My role: **Lead a team of seven AI engineers** in developing the AI components of our products, including the Learning Intelligent Management System (LIMS), Auto-Oral Defense, Unlimited Review Quiz System, Masterlytics, ThinkAid, and the Adaptive Extra-Credit Assignment System

EZRouting - (Blossoms AI Partnership) *Boston, Jan 2024 – Present*
AI Project Lead

- **Real-Time Bus/Route Matching System**
  - ∗ Developed a deep learning system to classify buses' GPS signal sequences into assigned routes in real time
  - ∗ Implemented deep learning interpretability mechanisms within the system to refine the model and improve dataset labeling, and to support anomaly detection with specific reasons to assist the school district in easier routing management
  - ∗ The product was **deployed in a Pennsylvania school district with over 100 schools**, and is planned for deployment to other school district customers
- **Intelligent Database Bot**
  - ∗ Designed a predefined computation system that supports AI-based input selection and function nesting to generate abstract analysis results, followed by report generation using an LLM based on computation results, preventing the LLM from generating inconsistent output or failing to handle computational tasks correctly
  - ∗ Integrated into EZRouting's app system, enabling consistent and error-free database analysis

Boston College *Boston, Aug 2023 – May 2024*
Teaching Assistant

- **Machine Learning (CSCI3345.01)**: Responsible for grading assignments, holding office hours, and designing certain homework and sample code
- **Vision and Learning (CSCI3399.01)**: Responsible for grading assignments, holding office hours, and hosting presentations

Publication

Leveraging LLMs and MLPs in Designing a Computer Science Placement Test System *Boston, 2023*
In Proceedings of CSCI 2023, Yi Li*, Riteng Zhang*, Danni Qu*, Maira Marques Samary

Talks

Understanding LSTM Networks *Boston, Nov 2023*
Boston College Experimental Math & ML lab

Travels

ACL 2023
NeurIPS 2023